UNIT 20: BUILDING A PERSONAL COMPUTER

ASSIGNMENT 2: PLANNING THE BUILD OF A PERSONAL COMPUTER

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BTEC First Diploma in Information and Creative Technology

Unit 20: Building a Personal Computer

Assignment 2 of 3 – Planning the Build of a Personal Computer - Hardware Installation Plan

Contents

Description of purpose and client requirements of the PC Build:

Our client is a local college, they have decided to add more computer systems within their Library as they have extra students this year, the purpose of the PC Build is to provide the college with low-cost additional desktops for their library to accommodate for their extra students.

The college has specified that the systems must meet the system requirements for Visual Studio Code, and have MS Office 365 and AVG Antivirus installed.

They have also noted that they do not have a lot of funds to spend, which we must keep the PCs low-cost as possible, even at the cost of longevity and performance, they only have to meet the requirements, however we should look into slightly better components for the alternative designs, as the college is open to spending more if we can justify the benefits. There is also a strict time limit where the PC must be built within 1 week; during the half term, as they cannot complete the work during term time.

We have been given a catalogue of the college's suppliers available components, along with the prices and delivery times, which we will use to select the components for the PC build.

For the PC build to meet its purpose and requirements we must make sure the system has the software specified installed, and meet the requirements for them, we should also take into account the prices and delivery times on the catalogue to make sure we meet the 1-week deadline and keep the cost as low as possible while ensuring compatibility.

Identify hardware components, software resources and tools required.

Hardware Components Required	Software Resources Required	Tools Required
Hard Drive	Windows 11	USB Drive
RAM	MS Office 365	Screwdrivers
CPU	AVG Antivirus	Thermal Paste
Motherboard	PassMark PerformanceTest	ESD Strap
Case	Component drivers	Anti-slip mat
Fans		
Power Supply		
Keyboard and mouse		
Monitor		

Description of required hardware components and software resources:

The hardware components that will be required for our system are the hard drive, RAM, CPU, motherboard, and PSU.

The hard drive is required because it stores the operating system, applications, and files. Without the hard drive there would be no storage method, making the computer unusable.

The RAM is used to store fast temporary data to run the operating system and applications, without the RAM the computer would not POST.

The CPU is the brain, it executes instructions to run software on your computer, such as your operating system, applications, it also plays a part in interfacing with your devices, without the CPU a computer would not be able to do anything and fail to POST, as there would be nothing to process instructions.

The motherboard is used to connect components together, such as making it possible for your CPU and RAM to communicate with each other, the motherboard also provides input and output ports so a user can interact with the PC, and it communicates with components installed to make them work fully, and to configure them. Without a motherboard, there would be no computer as there would be no functionality at all, as the components aren't going to be plugged into anything.

The PSU is used to provide components with safe and efficient power, the PSU will connect to components like the motherboard, hard drivers (SATA) the GPU (on power hungry models) or in fan hubs. The PSU also outputs different levels of power for some components as not every component will need high voltage, which could be largely inefficient, or sending high voltage into components that aren't designed for it can cause catastrophic failures. Without the PSU components would not be able to be powered, making the PC unusable.

Windows 11 is an operating system that runs on a computer, its purpose is to make it easy to run software and interface with the PC easily through a GUI, it simplifies the usage of the system as you won't need to remember or run multiple complex text only commands to do a simple task like creating a text file and printing it, which could be done with a few clicks in Windows 11's GUI, making it accessible to anyone.

MS Office 365 is a suite of productivity apps for typing documents, creating presentations, manging spreadsheets, creating posters, etc. For example, in education environment MS Office 365 would be used to create documents digitally instead of on physical paper, which makes it easy to change the content or create advanced documents that would be difficult or time consuming on a physical copy.

AVG Antivirus monitors the system it is installed on for malware to keep the user safe, its purpose is to scan for malware and block it from running, prevent damage to the system or user files.

PassMark Performance Test is a benchmarking tool that pushes the system components to their limits to test their performance and stability, which is essential in PC building as it helps diagnose any stability problems and if the components are performing as they should compared to the baseline of computers with the same specifications.

The drivers for your components are installed onto your operating system and they are required for getting the most out of your hardware, and are required in some aspects, they act as a translator for the OS and your hardware making it so they can communicate with each other and function properly, without your components drivers some hardware may not function or run with a fallback with a massive performance and feature loss. They are required for proper use of the system, no component drivers would make some programs not function correctly (such as Performance Test, and MS Office 365).

Compatibility Check of hardware components and software resources

S/W Required	**CPU**	**RAM** (MB)	**HDD Space** (MB)	**Graphics**	**OS**	**Costs**
PassMark PerformanceTest	**Model released in 2009 or later**	**4096MB**	**300MB**	**DirectX 9**	**Windows 7 x64**	**£0**
MS Office 365	**1.6GHz or faster**	**4096MB**	**4096MB**	**DirectX 9**	**Windows 11**	**Unknown, education MS Office 365 pricing is not displayed, and the college likely already has a license.**
AVG Antivirus	**Pentium 4 or Athlon 64 or above (supports SSE3)**	**1024MB**	**2048MB**	**N/A**	**Windows 7**	**£0 (for basic)**
Visual Studio Code	**1.6GHz or faster**	**1024MB**	**500MB**	**N/A**	**Windows 10**	**£0**
Op^{ing} System	**CPU**	**RAM** (MB)	**HDD Space** (MB)	**Graphics**		
Windows 11	**1GHz or faster with 2 or more cores and 64-bit.**	**4096MB**	**65536MB**	**DirectX 12 or later with WDDM 2.0 driver**		
Select highest spec OS from those above and fill in details for OS below	*Select highest spec CPU from above*	*Select largest RAM from above*	*Add all HDD space from above*	*Select highest spec graphics from above*		
Recommended Hardware Specifications	**1GHz or faster with 2 or more cores and 64-bit.**	*4096MB*	*72480MB*	**DirectX 12 or later with WDDM 2.0 driver**	**Windows 11**	

Hardware Specification Chosen

PC Model	**CPU**	**RAM (MB)**	**HDD Space (MB)**	**Graphics**	**OS**	**Costs**

Fujitsu D3923-A1	Intel i3 - LGA1200, 3.6 GHz Quad core, 6MB cache	8GB Crucial - 2400 MHz, DDR4	Patriot P320 - 128 gigabytes	iGPU from CPU	Windows 11	£320

Health and safety assessment:

Health and Safety Risks	Precautions Taken to Reduce the Risks
Untidy workstation	Tidying the workstation before using it, or changing where the work is taking place.
Busy/cramped environment	Clear the environment, if not possible move elsewhere or postpone.
Food and drink nearby electronics	Keep food and drink far away from any electronic device so if anything is spilled you will be safe.
Trip hazards (loose wires, random objects, uneven surface)	Move away loose wires and objects, same applies with uneven surfaces.
Electric shock from live components	Make sure the components are unplugged, and pay attention to warning labels on components you are handling or working near, as some components can hold dangerous amounts of charge, even when they have been unplugged for several hours, in some cases even days.
Not giving full attention	Ensuring that everybody involved is giving full attention to the task at hand, reducing the chance of risks occurring.
Damaged components	Safely and correctly disposing of the broken components, or having them professionally repaired if possible. Stopping all usage of the components to prevent further failure or safety risks
Jewellery	Remove or tuck away any loose jewellery to stop them getting stuck or touch live components.

Heavy equipment	Make sure those with back problems do not lift heavy equipment, having someone to help carry equipment would also reduce the risks for healthy people.

Functional Test plan (hardware components and software resources)

Functional Test	Expected Result
Does the CPU fit in the socket?	CPU fits into the socket and the cover is latched on fully
Check components and cables are fully connected.	All components and cables are fully seated, and that the components wiggle little as possible.
Check for missing screws	All screws needed are in their right place
Check if thermal paste is applied to the CPU	The CPU has thermal paste applied appropriately.
Is the power cable fully inserted in the power supply?	Power cable is fully pushed into the power supply socket
Is the PC case closed securely with no components exposed?	All case panels securely attached
Is the display cable inserted into a valid output source?	Display cable connected to GPU or motherboard (if iGPU) and outputs on boot
Does the RAM module fit in the slot?	The module fits into the slot without issues.

Does the PSU fit in the case, and do the cables fit our components?	PSU fits and can be screwed in, and the PSU cables work with our components.

Performance Test plan (hardware benchmark testing)

Functional Test	Expected Result
Run CPU benchmark with PassMark PerformanceTest	Benchmark completes without failing, and meets expected performance for the CPU.
Run Graphics benchmark with PassMark PerformanceTest	Benchmark completes without failing, and meets expected performance for the GPU.
Run Memory benchmark with PassMark PerformanceTest	Benchmark completes without failing, and meets expected performance for the RAM.
Run Disk benchmark with PassMark PerformanceTest	Benchmark completes without failing, and meets expected performance for the drive.
Check system temperatures during a full benchmark	Components don't overheat and the temperatures don't exceed safe limits and there is no thermal throttling.
Check system temperatures during idling	Components don't overheat and run relatively cool.
Check if system specifications are reported as expected	System info displays all the installed components, and are running at their expected specifications, example being correct RAM capacity and speeds.
Check if the required software runs appropriately for usage	Software runs with appropriate performance and does suffer with stability issues.

Does Windows 11 run stable and as expected?	Windows 11 runs without stability issues, and works within expected use case.

Description of 1st Alternative design idea:

This alternative PC design is slightly more tailored for longevity, it costs more because of this (totals at £360), however the better components mean it will be usable for much longer compared to the original PC, with the purpose of saving the college money in the long-term.

It would contain some slightly different components compared to the original design, it would have a “Patriot P320 256 gigabytes, 2200MB/s read speed, 1200MB/s write speed, NVME M.2” the upgrade to 256GB allows for room for extra programs to be installed in the future, and makes sure there’s room for programs to update, 128GB of storage is a very low minimum in 2026 as even simple programs can take up lots of space, and the operating system will also allocate some of that space.
For RAM we will be sticking with the “8GB Crucial 2400 MHz, DDR4, 8 GB”, because for the simple tasks the college has planned for 8GB of RAM should be enough for a few years, though there could be slowdowns with multiple tabs and applications open, but the cost for extra capacity is too great to look into for this alternative design idea.
Also with the CPU, we will be sticking with the “Intel i3 LGA1200, 3.6 GHz Quad core, 6MB cache” as it is perfectly fine for this use case and possibly even above, the office suite and other programs specified in the requirements are not resource intensive programs, which the i3 will be able to handle tasks from them quite quickly for many years to come.
We’re still staying away from a dedicated GPU, for the colleges use case a dedicated GPU is completely unnecessary, as the iGPU in the CPU is more than capable of handling the software on its own, as the Office suite, AVG Antivirus, and Visual Studio Code and not dependent on graphics, even for browsing in software such as Chrome with the iGPU it will provide a perfectly adequate user experience, having a dedicated GPU would be overkill for this design idea, and also would be inefficient, as the dedicated GPU would require extra power to run compared to the iGPU that is more than capable of the simple tasks.
For the motherboard we’re also sticking with the “Fujitsu D3923-A1" as its low cost and works with our components perfectly.

Description of 2nd alternative design idea:

This alternative design idea focuses purely on longevity, it costs a fair amount more than the cheapest as possible (totals at £415), however the better components mean it could be a perfectly usable system for at least 8 years, which means the college doesn't have to buy new PC's as often, saving them money in the long term.

It would contain a "Patriot P320 256 gigabytes, 2200MB/s read speed, 1200MB/s write speed, NVME M.2" which we also use in the 1st alternative design, which we go over the reasons why we chose this in the 1st alternative design, as it still applies. For this design with the RAM, I decided to go with the "16GB Samsung 3200 MHz, DDR4" module, because 8GBs of RAM in 2026 can be regarded as low for multitasking, which a student will likely be doing, browsers can consume large amounts of memory, which only increases with every new tab, and operating systems and software as they update also become more demanding for resources. For the CPU we're still sticking with the "Intel i3 LGA1200, 3.6 GHz Quad core, 6MB Cache" as it is still a perfectly fine processor for many tasks, while it isn't incredibly powerful, for the basic tasks that will be done on these PC's it will do the job for many years to come. Again, like in the previous designs we are not using a GPU and we are relying on the iGPU, as the GPUs will be overkill for these systems, and the iGPU will handle the tasks fine. For the motherboard we're again sticking with the "Fujitsu D3923-A1" because of its reasonable pricing, it still supports our RAMs specifications, supports integrated graphics, and M.2.

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Assignment 2 of 3 – Planning the Build of a Personal Computer - Hardware Installation Plan

Compatibility Check of hardware components and software resources for 1st alternative design idea

S/W Required	CPU	RAM (MB)	HDD Space (MB)	Graphics	OS	Costs
PassMark PerformanceTest	**Model released in 2009 or later**	**4096MB**	**300MB**	**DirectX 9**	**Windows 7 x64**	**£0**
MS Office 365	**1.6GHz or faster**	**4096MB**	**4096MB**	**DirectX 9**	**Windows 11**	**Unknown, education MS Office 365 pricing is not displayed, and the college likely already has a license.**
AVG Antivirus	**Pentium 4 or Athlon 64 or above (supports SSE3)**	**1024MB**	**2048MB**	**N/A**	**Windows 7**	**£0 (for basic)**
Visual Studio Code	**1.6GHz or faster**	**1024MB**	**500MB**	**N/A**	**Windows 10**	**£0**
Op^{ing} System	**CPU**	**RAM** (MB)	**HDD Space** (MB)	**Graphics**		
Windows 11	**1GHz or faster with 2 or more cores and 64-bit.**	**4096MB**	**65536MB**	**DirectX 12 or later with WDDM 2.0 driver**		
Select highest spec OS from those above and fill in details for OS below	***Select highest spec CPU from above***	***Select largest RAM from above***	***Add all HDD space from above***	***Select highest spec graphics from above***		
Recommended Hardware Specifications	*1GHz or faster with 2 ore more cores and 64-bit.*	*4096MB*	*72480MB*	**DirectX 12 or later with WDDM 2.0 driver**	**Windows 11**	

Hardware Specification Chosen

PC Model	CPU	RAM (MB)	HDD Space (MB)	Graphics	OS	Costs

Fujitsu D3923-A1	Intel i3 - LGA1200, 3.6 GHz Quad core, 6MB cache	8GB Crucial - 2400 MHz, DDR4	Patriot P320 - 256 gigabytes	iGPU from CPU	Windows 11	£360

Compatibility Check of hardware components and software resources for 2nd alternative design idea

S/W Required	**CPU**	**RAM** (MB)	**HDD Space** (MB)	**Graphics**	**OS**	**Costs**
PassMark PerformanceTest	**Model released in 2009 or later**	**4096MB**	**300MB**	**DirectX 9**	**Windows 7 x64**	**£0**
MS Office 365	**1.6GHz or faster**	**4096MB**	**4096MB**	**DirectX 9**	**Windows 11**	**Unknown, education MS Office 365 pricing is not displayed, and the college likely already has a license.**
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Oping System	**CPU**	**RAM** (MB)	**HDD Space** (MB)	**Graphics**		
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Select highest spec OS from those above and fill in details for OS below	***Select highest spec CPU from above***	***Select largest RAM from above***	***Add all HDD space from above***	***Select highest spec graphics from above***		

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Recommended Hardware Specifications	*1GHz or faster with 2 or more cores and 64-bit.*	*4096MB*	*72480MB*	**DirectX 12 or later with WDDM 2.0 driver**	**Windows 11**	

Hardware Specification Chosen

PC Model	CPU	RAM (MB)	HDD Space (MB)	Graphics	OS	Costs
Fujitsu D3923-A1	Intel i3 - LGA1200, 3.6 GHz Quad core, 6MB cache	16GB Samsung - 3200 MHz, DDR4	Patriot P320 - 256 gigabytes	iGPU from CPU	Windows 11	£415

Identify the installation sequence to build the computer system.

Planning & Preparation	Installation	Testing and Completion
Clear workspace of clutter	Remove case side panel	Place the side panel back on, plug in the PC and power it on and make sure it boots to the BIOS
Wear anti-static wrist strap	Slot in PSU	Install Windows and the drivers
Make sure components are set on a non-slippery surface	Place motherboard and screw it in	Run benchmarking and review the points and see if it matches the baseline, and that the system ran stable
Move away food and drink	Insert CPU	Install the software specified in the requirements
Determine what screwdriver you will need	Place CPU cooler	Run the software and see if they work as expected
Prepare USB flash drive to contain the Windows 11 installer	Insert RAM	Monitor component temperatures to see if they are within safe ranges
Check if components come with thermal paste, if not purchase some	Insert M.2 Drive	Check system specifications to see if they match what is expected
Read through component manuals if given	Connect PSU to motherboard	
Make sure components appear in good shape and are the correct models	Wire up case to motherboard	

Explanation of installation sequence to build the computer system:

Place the case and components onto a non-slippery surface, and if possible, place them on an anti-static mat or material. Put on an anti-static wrist strap onto your wrist and clip it to the metal on your case or any non-live metal. Make sure the environment is clear of hazards such as food and drink, and that the desk and surroundings are clear of clutter.

Remove the case side panel by sliding it off, you might need to remove a screw on the panel itself, then slot in the PSU to its designated place and put in the screws on the case to secure the PSU.

Place the motherboard into the case and refer to your motherboard case manual to figure out where your screws should go, and then screw them in.

Prepare to insert the CPU by lifting the latch, removing the cover, and place the CPU carefully into the socket aligning the golden triangle on the corner of the CPU with the one on the socket, and then push the latch back down to lock the CPU in place. Apply the thermal paste to CPU in a pea sized amount in the centre, then place the CPU cooler onto the socket making sure the screw posts line up the with the motherboards and then screw it in fully, not being fully screwed in could cause poor contact with the CPU and the cooler causing high temps.

Insert the RAM into its slot by aligning the notch on your RAM with the bridge in the slot, then press down carefully with equal pressure on the module until it clicks into place.

Then insert the M.2 NVME drive into the slot on the motherboard at a slight upwards angle, and gently wiggle the drive sideways until the drive fully inserts into the slot, then screw it in.

Finally, connect the wires from the PSU to the motherboard, the wrong connectors won’t fit in the wrong sockets, you also won’t need to push with excessive force, it’s probably the wrong socket if you feel like you need to do that.

Then connect any extra cables you might have (such as your cases front panel) to their places.

Justification of original design idea; how hardware components and software resources meet the purpose and client requirements:

The original design idea matches the colleges requirements and meets its purpose, the college needs new desktops that are low cost as possible to accommodate for the extra students, the college does not have lots of funding to splash out on the desktops. Our PC meets the requirements because I have chosen the cheapest components possible within the list we were provided, however you may notice I did choose the 128GB M.2 drive instead of the cheap HDD, this is because the motherboard we chose on the list does not support IDE, which would make the IDE hard drive useless and a waste of money, which would make us fail to provide a functional system for the college. The systems components while they aren't high specification, they meet the software system requirements specified in the scenario, which should provide an adequate experience for the new college students, all while maintaining the colleges preference for low cost as possible. This design fulfils the purpose the college wants, the system will be able to run the Office suite, AVG Antivirus, and Visual Studio Code, as they expect. This makes this design the best option for matching the colleges requirements strictly, as it meets the time limits, low cost, and their software needs are met.

Justification of 1st alternative design idea; how hardware components and software resources meet the purpose and client requirements:

This alternative design is very similar to the original design, it still fits its designated purpose for the new students, however this alternative design costs slightly more, while the requirements did specify that the PCs should be low cost as possible, the college did note that they are open to spending more if the benefits are justified, this design idea contains the almost the same specifications as the original design, except it uses the "Patriot P320 - 256 gigabytes" M.2 drive instead of the 128GB variant, this was chosen because it would be very beneficial to the college, the bigger drive would reduce maintenance costs as half of the 128GB drives space would be taken up by the applications and OS, which would become a problem as updates will increase the software sizes, and even though its likely students will have their work backed up to the cloud, some of their work would be saved on the PC and some of the software may cache data for the user of various sizes, which if multiple students use the PC it could easily be filled up, causing data loss or system instability. The 256GB drive would leave plenty of space for extra software, OS and software updates, local student files, which this would also help longevity of the systems, potentially saving the college money in the long term. This design if the college was open to spending slightly more with these benefits would be the best option for future-proofing the desktops, while still keeping the costs low.

Justification of 2nd alternative design idea; how hardware components and software resources meet the purpose and client requirements:

This alternative design is like the 1st alternative design, it still fits the college's purpose, though this design focuses on better usability and futureproofing of this system, the cost is steeper than the original design, but the college should consider the benefits of this designs focus, instead of low-cost as possible. This design keeps the 256GB M.2 drive from the 1st alternative design, as it would be enough for the tasks done on the PC, and leaves room for adding more, however this design uses the "16GB Samsung – 3200MHz, DDR4" RAM module, the 2x capacity would improve productivity with the students as the extra RAM capacity keeps multitasking smooth, 8GBs of RAM would work for light multitasking, but it would soon start to fill and make the system crawl and unstable, an example where there would be noticeable problems for the system is if the user had lots of tabs open at once, visual studio code running, multiple Office programs with large documents and slides, which then the 8GB could be fully used up, which could cause the desktops to freeze or crash, leading to unsaved work being lost, or possibly even corruption. Which this could become more of a problem as the systems age, as the programs would become more demanding. The upgrade to 16GB would solve this problem for many years, and the chance of a student filling up the 16GBs of RAM from the basic tasks is incredibly unlikely, this would also save the college money in maintenance costs and from not needing to purchase replacements as often, as the systems stay usable for longer. This design would be a good choice for the college if they were open to spending more, it's better for productivity, and these specifications could keep it running for many years.

DERBY college

Explanation of constraints and measures that can be taken to resolve constraints:

The college had given us many constraints when designing the systems, they specified it must be built within 1 week as they cannot complete this work during term time, meaning there was a strict time limit that had to be considered when deciding on the components, luckily the components that were chosen met this time frame. Regarding the components the selection of components were very constraining, some would not be delivered within this time frame, or far too expensive, some completely incompatible with the hardware, the catalogue we were given also had very poor selection; a very similar desktop could be produced for a lower price if the college changed their supplier, with the catalogue constraints and request of lowest price as possible, the desktops may age poorly, such as going slow, and the storage getting constantly filled. To resolve all of these constraints, the college should look into changing their suppliers for better prices and options, and they should look into the benefits of the alternative designs and their futureproofing, as it might make it possible for them to spend more on the components later on from less replacements.

Justification of chosen design idea, including hardware components and software resources and why it may be best suited compared with the other ideas to meet the purpose and client requirements:

I'm choosing the original design idea as its best suited to the college's requirements, they specified that the cost must be kept as low as possible which the 1st design follows, while the other designs may be better for the long term it is the college's decision on spending more. The original design composes of the cheapest compatible components for the software and hardware, the specifications of the hardware meet the requirements for Windows 11, MS Office, AVG Antivirus, and Visual Studio Code, the hardware also meets the requirements for the college as they are the cheapest components that were compatible with each other, and fit in the strict time frame. This PC fits the colleges requirements as described, and the design will be able to fulfil its purpose for college, as the components exceed the minimum requirements for the software.